The composite Trapezoidal Method

$$\int_0^4 e^x\,dx = \int_0^1 e^x\,dx + \int_1^2 e^x\,dx + \int_2^3 e^x\,dx + \int_3^4 e^x\,dx$$

$$Let\ us\ apply\ Trapezoidal\ Method\ for\ each\ integral$$

$$\int_0^4 e^x\,dx \cong \frac{1-0}{2}[f(0)+f(1)] + \frac{2-1}{2}[f(1)+f(2)] + \frac{3-2}{2}[f(2)+f(3)] + \frac{4-3}{2}[f(3) + f(4)]$$

$$I = \frac{1}{2}[f(0)+2f(1)+2f(2)+2f(3)+f(4)]$$

This is called composite Trapezoidal Method

In genaral form we can write:

$$\int_a^b f(x)dx,\ \ h = \frac{b-a}{n} \quad where\ n\ is\ a\ positive\ inetger\ \ greater\ than\ zero$$

$$I_T = \frac{h}{2}[f(a)+2f(a+h)+2f(a+2h)+\cdots+2f(a+(n-1)h)+f(b)]$$

The composite Simpons' Method

$$\int_0^4 e^x\,dx = \int_0^1 e^x\,dx + \int_1^2 e^x\,dx + \int_2^3 e^x\,dx + \int_3^4 e^x\,dx$$

$$Let\ us\ apply\ Trapezoidal\ Method\ for\ each\ integral\ \ h = \frac{b-a}{2}$$

$$\int_0^4 e^x\,dx \cong \frac{1-0}{3*2}[f(0)+4f(0.5)+f(1)] + \frac{2-1}{3*2}[f(1)+4f(1.5)+f(2)] + \frac{3-2}{3*2}[f(2)+4f(2.5)+\ f(3)] + \frac{4-3}{3*2}[f(3)+4f(3.5)+f(4)]$$

$$I_s = \frac{1}{3*2}[f(0)+4f(0.5)+2f(1)+4f(1.5)+2f(2)+4f(2.5)+2f(3)+4f(3.5)+f(4)]$$

$$in\ general\ form\ we\ can\ write\ the\ composite\ Simpson's\ method:$$

$$\int_a^b f(x)dx,\ \ h = \frac{b-a}{n} \quad where\ n\ is\ an\ even\ positive\ inetger\ \ greater\ than\ zero$$

$$I_S = \frac{h}{3}[f(a)+4f(a+h)+2f(a+2h)+\cdots+2f(a+(n-2)h)+4f(a+(n-1)h)+f(b)]$$

Example:

Solve the following integral with taking 11 samples point by Trapezoidal Method.

$$\int_1^6 (2+\sin(2\sqrt{x})\,dx$$

$$With\ saying\ 11\ samples\ we\ mean\ the\ points\ are\ x_0, x_1, \dots., x_{10}$$

$$h = \frac{6-1}{10} = 0.5$$

$$I_T = \frac{0.5}{2}[f(1) + 2f(1.5) + 2f(2) + 2f(2.5) + 2f(3) + 2f(3.5) + 2f(4) + 2f(4.5) + 2f(5) + 2f(5.5) + f(6)]$$

$$I_T = \frac{0.5}{2}[2.909 + 2 * 2.638 + 2 * 2.308 + 2 * 1.979 + 2 * 1.683 + 2 * 1.435 + 2 * 1.243 + 2 * 1.108 + 2 * 1.028 + 2 * 1 + 1.017] = 0.5 * \frac{30.77}{2} = 8.193 ????$$